ANSWERS

***High** voltage pn diodes.*

a) Explain the process by which carriers can move from the junction region of a diode to the contacts under forward bias. Define in terms of the relevant parameters of the diode that control that process what is meant by the long and short diode approximation. *[4]*

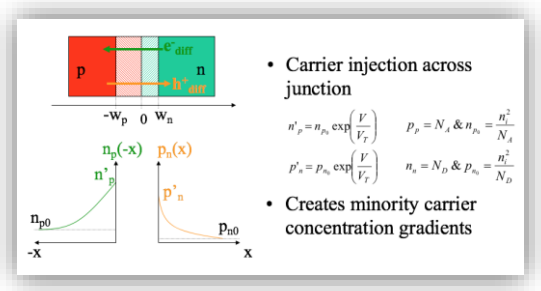
[Basic concepts, but asked in a slightly different way than presented in the lectures]

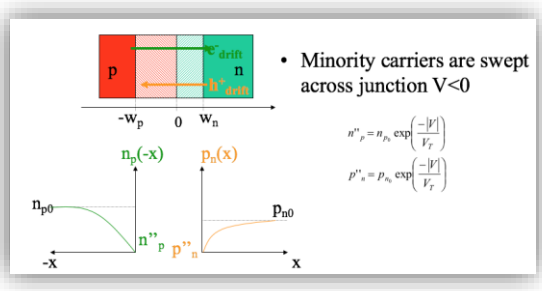
Diffusion dominates as minimal electric field outside of depletion region, diffusion constant, *D*, the relevant parameter. **Recombination** means that the minority carriers are progressively lost as they progress to the contact, with the number dependent on the recombination rate, or its reciprocal diffusion time, τ. Majority carriers flow in from the contacts to supply the recombination.

The average distance a carrier travels before it recombines, the **diffusion length**, is given by $L = \sqrt{(D\ \tau)}$. The long diode approximation is the limit when the distance to the contact is much **longer**, and the short diode is when the distance to the contact is much **shorter** than *L*.

b) Draw two diagrams showing the variation of the excess minority carrier density along the entire length of a diode under forward and reverse bias making no approximations. A p^+n diode has doping of [p] = 10^{16} cm^{-3} and [n] = 10^{14} cm^{-3}. Calculate the values of the excess minority hole carrier concentration at the edges of the depletion region and at the contacts for a forward and reverse bias of 1 V. *[6]*

[application of knowledge and calculation],





Diagram, 2 marks each. Note slopes are not zero at contact in general case. Lose one mark if ignored.

Concentrations – see above for formulas:

V = 2 V, so exponential multiplier is exp(2/.026) = 5E16, from 0.026 V given in formula sheet.

n_i given as 1.45E10 /cc in formula sheet, ni^2 = 2.1E20 /cc^2

Forward bias

p_{n0} = 2.1E20/1E14 = 2.1E6 /cc, but dn_{po}, excess = 0 [1]

p'_n = 5E16*2.1E6 = 10.5E22 and dp'_n is practically the same [1]

Reverse bias

p_{n0} = 0, as before [1]

p''_n = 2.1E6/5E16 = 4E-11/cc, so $dp''_n = -p_{n0}$ = -2.1E6/cc [1]

c) *The steady-state current flow in a diode can be derived by looking either at the current flow at the edge of the depletion region or the charge needed to be resupplied to maintain a steady state. From equating these two approaches show that the excess minority carrier concentration, δp_n in the p region of the diode will be given by solving*

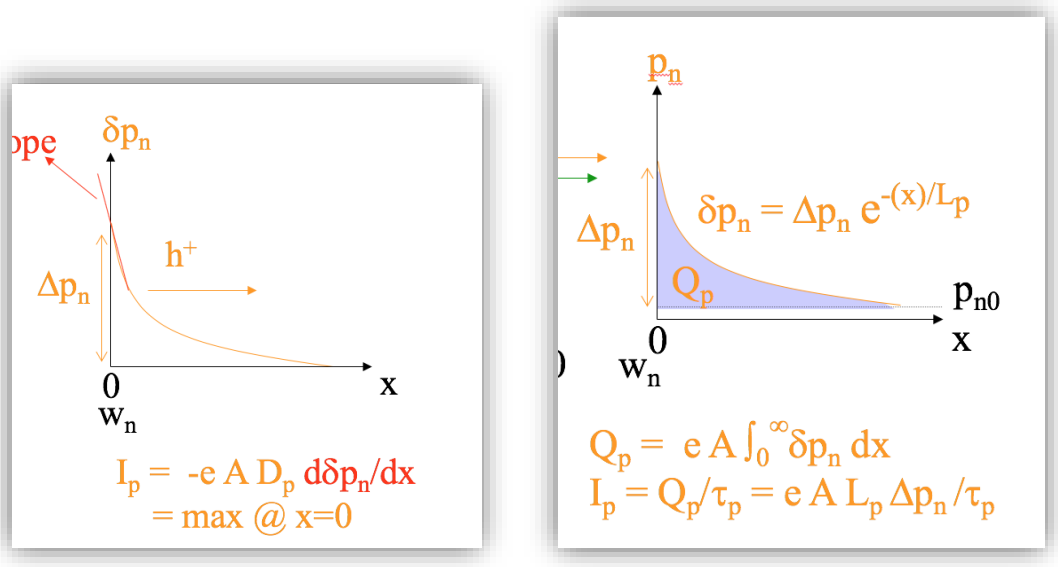
$$\delta p_n = D_p \tau_p \frac{d^2 \delta p_n}{dx^2}$$

where D_p is the diffusion constant and τ_p the recombination time. Solve this equation for δp_n using appropriate boundary conditions for the long-diode approximation.

Why is it not possible to use this equation to solve for the short-diode approximation? *[14]*

[initial bookwork followed by original derivation and consideration of applicability]

The two relevant formulas are given below from the lecture notes:



[2]

Hence we can equate

$$D_p \frac{d\delta p_n}{dx} = \frac{1}{\tau_p} \int_0^{contact} \delta p_n dx$$ [2]

Differentiating and rearranging:

$$\delta p_n = \tau_p D_p \frac{d^2 \delta p_n}{dx^2}$$ [2]

as required.

Solving by integration:

$$\delta p_n = \tau_p D_p \frac{d^2 \delta p_n}{dx^2}$$

This has the general solution

$$\delta p_n = Ae^{x/L} + Be^{x-/L} + Cx + D$$ [2]

where $L = \sqrt{\tau_p D_p}$.

when $x = 0, \delta p_n = \delta p_{n0}$ giving $\delta p_n = A + B + D$.

δp_n= 0 at the contact due to the unlimited number of carriers available and we then apply the long diode approximation, that the contact is much further away than $L = \sqrt{\tau_p D_p}$.

Hence $A = 0$ to ensure δp_n remains finite and $C = D = 0$ giving $\delta p_n = \delta p_{n0} e^{x/L}$. [2]

This solution is constructed by consideration of the recombination of the excess minority carriers. In the short diode approximation such a recombination does not occur, so it is not possible to formulate the solution in this way. [2]

2. *Switching delays*

a) *What is the dominant recombination process of carriers in silicon semiconductors under low electric fields and why are other processes slower? How does a high field contribute additional recombination?* *[6]*

[Bookwork but not as directly presented in lectures]

Shockley-Read-Hall recombination via defect states in the bandgap as SRH [1]

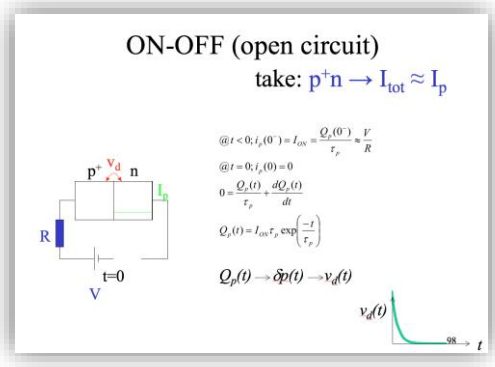
1) Does not require a direct band gap which significantly reduces direct optical recombination [1]
2) Is more likely than thermal excitation across full bandgap due to exponential dependence on the energy difference [1]
3) Requires no electric field to provide additional kinetic energy [1]
4) Does not rely on higher energy states above the conduction band edge that are only likely under heavy doping, such as Auger. [1]

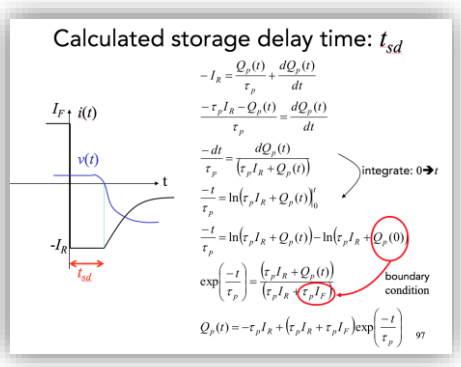
Under high electric fields carriers can be accelerated to provide sufficient energy to ionise on collision, provide additional excess carriers for recombination and therefore increasing the recombination rate.

b) *A diode in series with a resistive load is switched off in two ways. In the first, the forward bias is removed, in the second it is reversed. Sketch the variation of the excess hole charge after the switching occurs in both cases. Prove that when the excess hole charge is removed in the second case (reverse switching), half the initial charge still remains in the first case.* [14]

[Initial bookwork leading to solution of novel problem]

From the lecture notes, need to look at the solutions of the equation:





[4]

If $I_F = I_R = I$ then for the first case,

$$Q_1(t) = I\tau_p e^{-\frac{t}{\tau_p}}$$

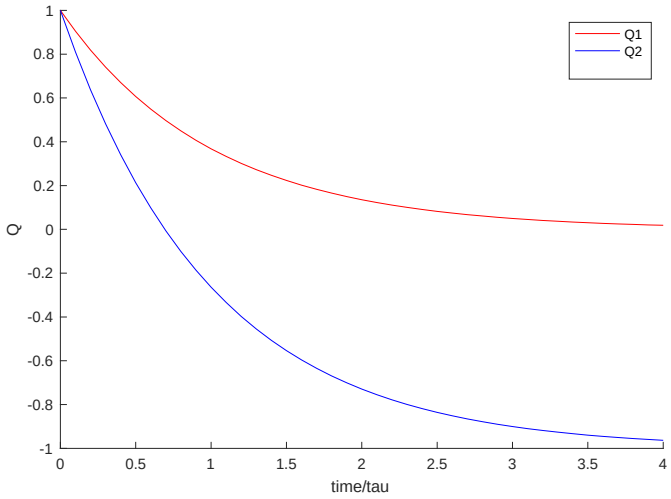
[1]

For the second case,

$$Q_2(t) = I\tau_p(2e^{-\frac{t}{\tau_p}} - 1)$$

[1]

These plot up to give:



[4]

When $Q_2(t) = 0$, $e^{-\frac{t}{\tau_p}} = \frac{1}{2}$, giving $\frac{t}{\tau_p} = \ln 2$. [2]

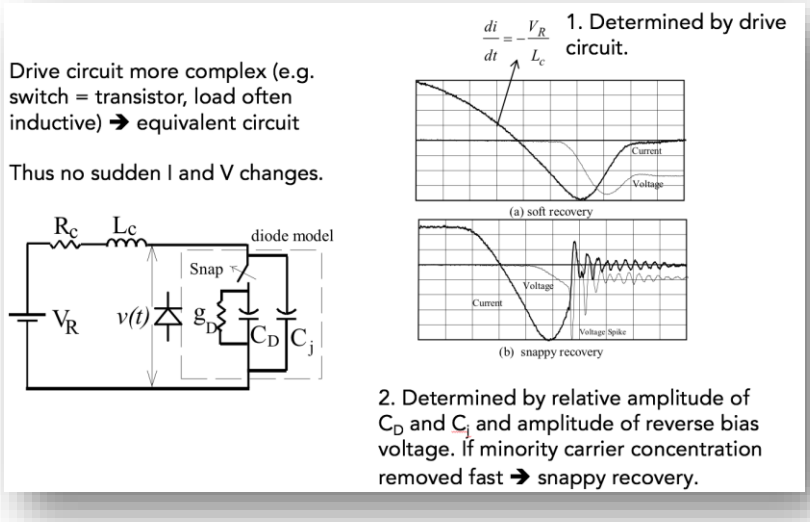
Substituting into $Q_1(t) = I\tau_p e^{-\frac{t}{\tau_p}}$ gives $Q_1(t) = \frac{I\tau_p}{2}$, which is half the value of $Q_1(0)$. [2]

(c) *What will cause additional delay in the second case before the current is completely switched off in both cases? Sketch the variation of current in a real example of diode switching, noting the cause of any differences from the ideal behaviour.* *[5]*

[Bookwork and application]

The additional delay will be due to the junction capacitance that will prevent the current returning to zero as soon as the excess charge is removed. This capacitance is due to the charge separated across the depletion region. With the load resistor, this produces an RC time constant that determines how fast the current returns to zero. [3]

Examples, from notes:



[2]